

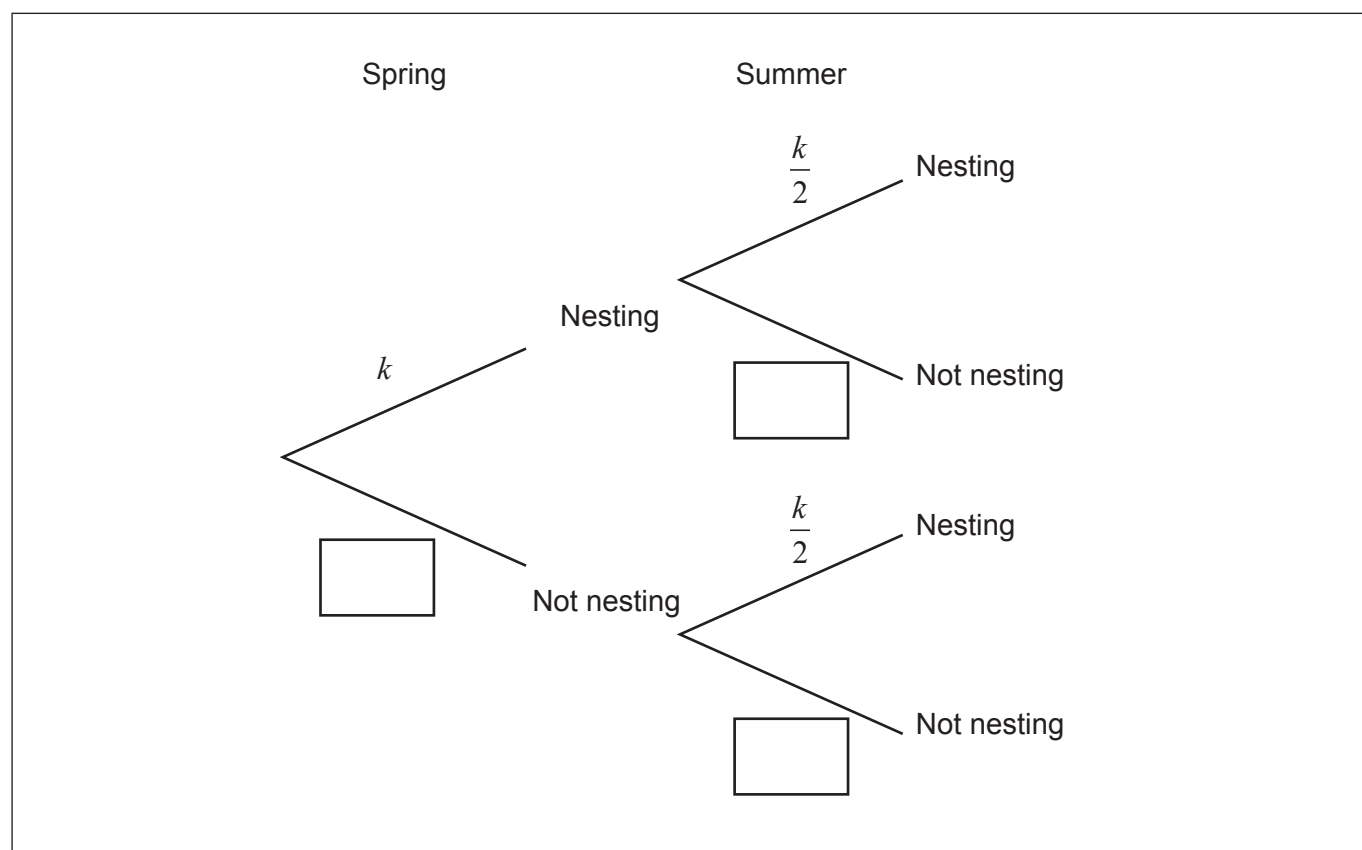
4. [Maximum mark: 6]

A species of bird can nest in two seasons: Spring and Summer.

The probability of nesting in Spring is  $k$ .

The probability of nesting in Summer is  $\frac{k}{2}$ .

This is shown in the following tree diagram.



- (a) Complete the tree diagram to show the probabilities of not nesting in each season. Write your answers in terms of  $k$ .

[2]

It is known that the probability of not nesting in Spring and not nesting in Summer is  $\frac{5}{9}$ .

- (b) (i) Show that  $9k^2 - 27k + 8 = 0$ .

- (ii) Both  $k = \frac{1}{3}$  and  $k = \frac{8}{3}$  satisfy  $9k^2 - 27k + 8 = 0$ .

State why  $k = \frac{1}{3}$  is the only valid solution.

[4]

(This question continues on the following page)



(Question 4 continued)

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